

Inferring mean wavelength
from DCR 1st moments.

Creating an estimator function $\langle \lambda \rangle = f(\Delta R_0)$...

Getting an analytic function is tricky.

Issue 1: This is the expression for R_0 as a function of wavelength (Kosakowski 2009):

$$R_0 = \frac{n^2 - 1}{2n^2} \quad (2)$$

Where n is described by (with some additional factors and terms if considering atmospheric effects):

$$[n(\lambda) - 1]10^6 = 64.328 + \frac{29498.1}{146 - (1/\lambda)^2} + \frac{255.4}{41 - (1/\lambda)^2}, \quad (3)$$

This is not a trivially invertible function (albeit I have yet to really try).

Creating an estimator function...

Getting an analytic function is tricky.

Issue 2: Because the R is not linear in wavelength, in general

$$\langle R \rangle \neq R(\langle \lambda \rangle)$$

We can expand to estimate the mean R with the mean wavelength but this further complicates the function that must be inverted and requires assumptions about the SED.

$$\left| \langle R \rangle \approx R(\langle \lambda \rangle) + \frac{1}{2} R''(\langle \lambda \rangle) \text{Var}(\lambda) + O(\text{3rd moment}) \right|$$

*Kosakowski states (without reasoning unfortunately), that “it is sufficient to know the effective wavelength of the object within a given bandpass” to get the mean R , implying that the correction is small.

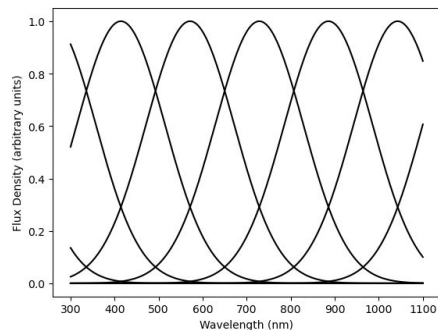
Creating an estimator function...

One simple non-analytic solution:

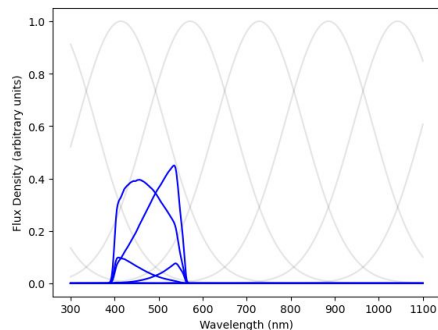
1. Create a set of known ΔR_0 and $\langle \lambda \rangle$ (mean wavelengths of the filtered SEDs).
2. Use the set to interpolate the mean wavelength from the measured DCR.

Created an interpolation set by using “template SEDs” with well defined means that span the wavelength range (gaussians in this case).

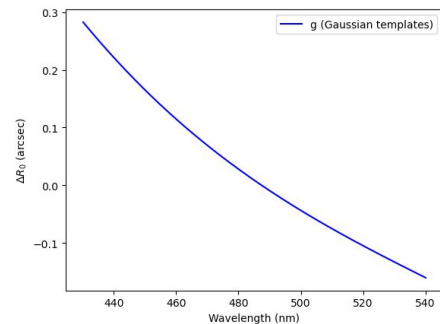
Note: the width of the gaussians appears to matter a lot here. Std between 100 and 150 nm seems to work well.



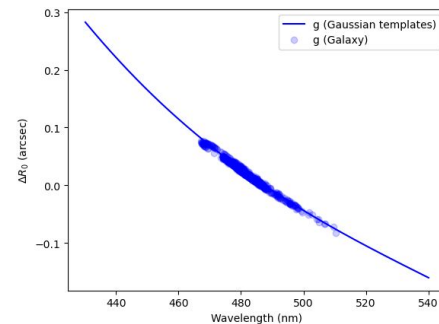
Create the template set (just 8 of the 5000 plotted here)...



Filter each template and calculate mean R and λ (g band shown here)



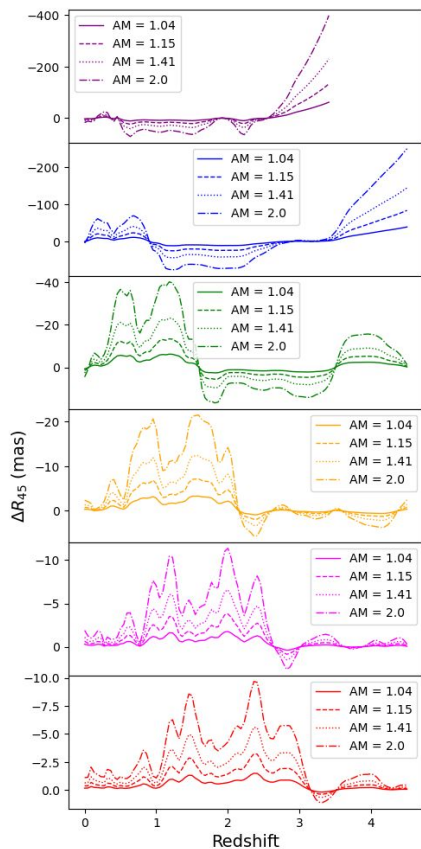
Plot (mean R - reference R) against mean λ . This is the interpolation set.



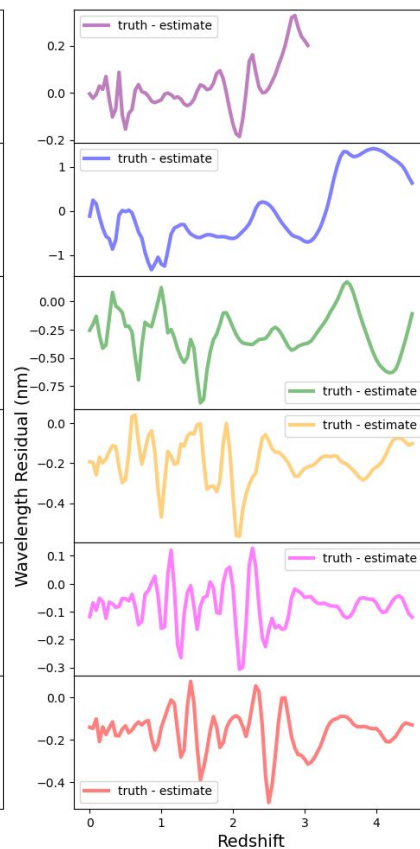
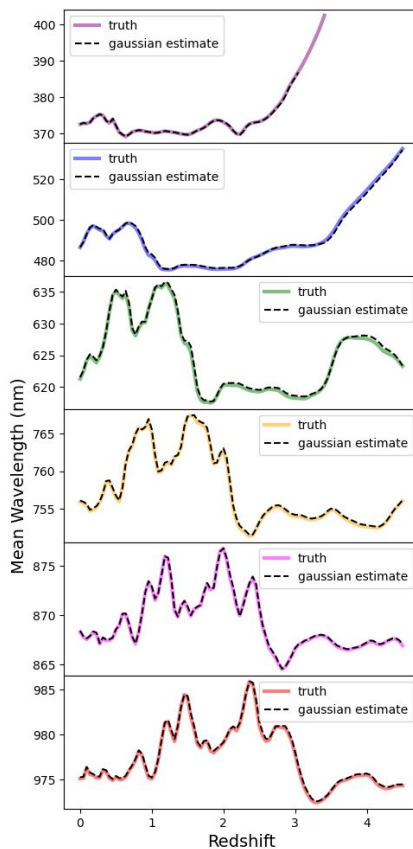
Check against some of the galaxy SED templates I've been using. Looks good!

The gaussian interpolation method appears to work well, recovering the mean wavelength within $\sim 1\text{nm}$ accuracy (given exact ΔR_0).

This plot shows a single Galaxy SED template (S0_A_0.sed) that has been redshifted by different amounts then the DCR and mean wavelengths calculated for each band.



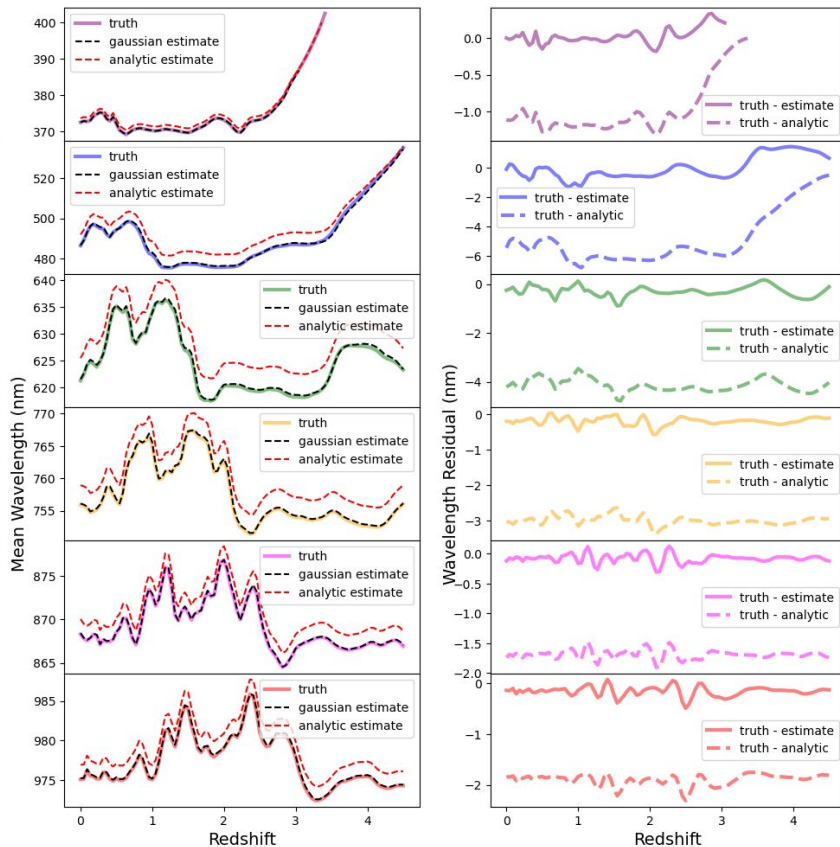
Note the inverted y axis



The residuals are small but have some offset from 0 that seems to be related to the choice of width for the gaussian templates.

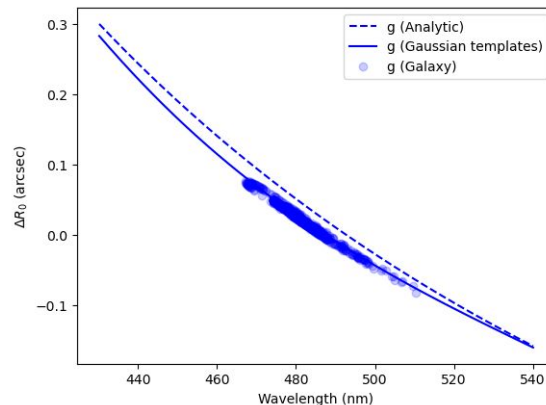
What if you assume $\langle R \rangle \approx R(\langle \lambda \rangle)$ like Kosakowski (essentially ignoring any effects from the width of the SED)?

Calling this the “analytic estimate” but I still had to do some interpolation here because inverting R is hard...



Much larger residuals with this assumption but still within $\sim 5\text{nm}$.

Can also see this method doesn't work well with the set of Galaxy SEDs from before:



What about the raw mean wavelength in the band (before filtering)?

The scatter is larger but it looks like you can still get a rough estimate. Although maybe we don't really care about this value anyways.

